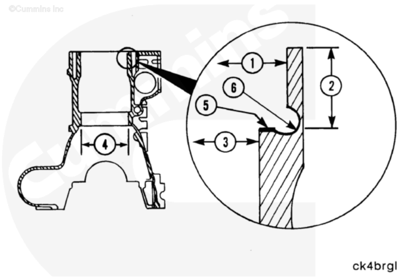
Trainings

General Information

There are seven variations of K19 cylinder blocks and oversize liners now in the field. Use the following information to identify the parts that are used. Make sure the correct parts are used with the correct block.

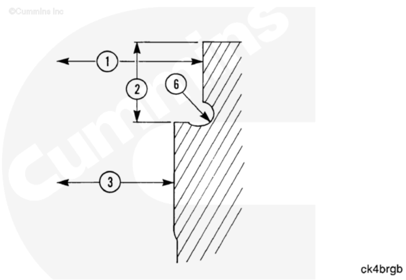
Definition of counterbore features:

- (1) Upper counterbore inside diameter, called upper press fit inside diameter
- (2) Counterbore depth
- (3) Lower counterbore inside diameter, called lower press fit inside diameter
- (4) Packing ring bore
- (5) Counterbore ledge
- (6) Counter bore radius (graphic illustrates double undercut radius).



The thick flanged block:

- (1) The upper press fit inside diameter of the bore is machined at the factory and referred to as standard. The inside diameter of the standard bore is 188.16 to 188.21 mm [7.408 to 7.410 in].
- (2) The counterbore standard depth for some engines built between September 1994, engine serial number 37155415, and June 1995, engine serial number 37158125, and after is 13.746 to 13.818 mm [0.5412 to 0.544 in]. The counterbore standard depth for all engines built June 1995, engine serial number 37158125, is 13.754 to 13.818 mm [0.5422 to 0.544 in]. The counterbore standard depth for all blocks manufactured from February 1981 to September 1994, engine serial number 37155415, is 13.23 to 13.28 mm [0.521 to 0.523 in].
- (3) Lower press fit inside diameter of the bore is machined at the factory and referred to as standard. The



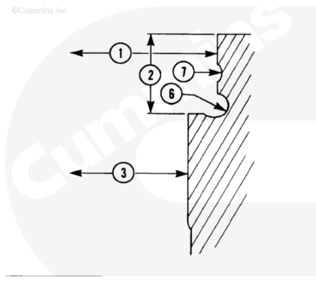
inside diameter of the standard bore is 180.09 to 180.14 mm [7.090 to 7.092 in].
- (6) The thick flange block has the double undercut radius.

Most engines with an engine serial number greater than 311216 70, February 1981, contain this type of block.

All K19 engines with engine serial number that begins with 37, December 1986, and engines with a serial number that begins with a 66, have this type of block.

The thin flange factory modified block is the same as the thick flange block except for the groove (7), which has **no** function.

This block is treated exactly the same as the thick flange block.



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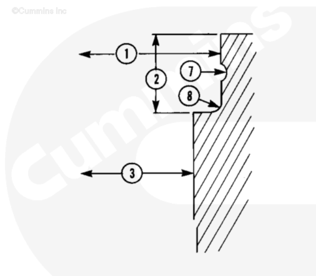
The thin flange field modified to thick flange block will be referred to as the ex-thin flange block.

- (1) The upper press fit inside diameter varies with the upper press fit oversized liner used during the modification.
- (2) The depth is 13.23 to 13.28 mm [0.521 to 0.523 in].
- (3) The lower press fit inside diameter is the same as the thick flange block.
- (8) It has a 1.73 to 1.85 mm [0.068 to 0.073 in] radius that is **not** undercut.

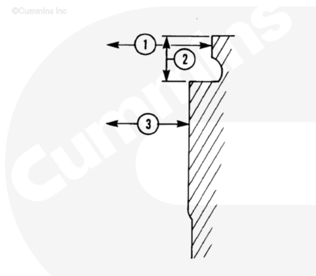
This type block is found on engines with engine serial number between 31103629, November 1976, and 31121670, February 1981. as they were originally built with thin flange blocks.



ck4brgi

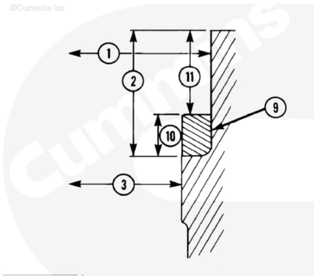
The thin flange block has:

- (1) A upper press fit inside diameter that can vary depending on the type of the upper press fit oversized thin flange liner used. Standard thin flanged liners are available. Oversized thin flange liners are no longer available.
- (2) Counterbore depth is 7.62 to 7.67 mm [0.300 ot 0.302 in].
- (3) Lower press fit inside diameter is the same as the thick flange type.



The radius is double undercut. This type of block is found on engines with an engine serial number range of 31103629 to 31121669.

The counterbore ring thick flange (8) is identified by the counter bore ring.



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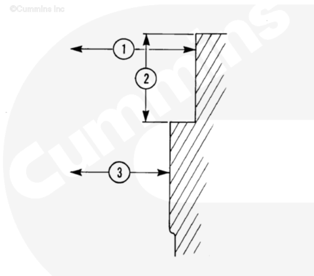
Depth (2), depth (11), and the inside diameter (3) are the same as present K38 and K50 engines, and thus use standard and oversized K38 and K50 liners. This type is **not** lower press fit. That is, by design there was **not** always a press fit below the liner flange (3). There is no recommended salvage for the inside diameter (3) on these blocks.

This type block is found on engines with an engine serial range between 31101150, August 1975, to 31103628, November 1976.

The original K19:



ck4brgg

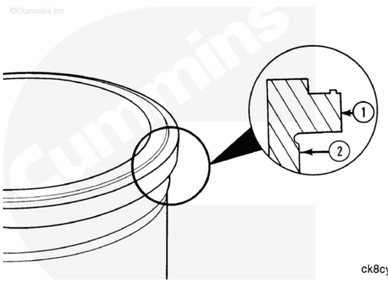
- (1) Upper press fit inside diameter is the same as the present K38 and K50 engines.
- (2) Depth is the same as the present K38 and K50 engines.
- (3) Lower counterbore is the same as the present K38 and K50, but will probably **not** provide press fit to the liner.

There is no recommended salvage for the inside diameter (3) on this type block.

This type of block is found on engines with an engine serial number range of 31100101, July 1974, to 31101149, August 1975.

Cummins Inc. recommends this style block **not** be reused.

The cylinder liners are:



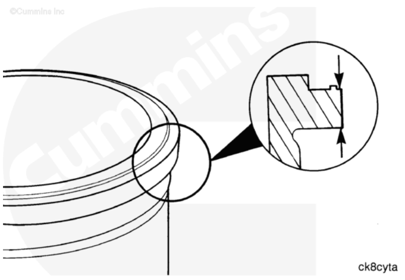
ck8cytc

- (1) Flange outside diameter referred to as the upper press fit outside diameter.
- (2) Referred to as lower press fit outside diameter.

The standard K19 liners are those liners used in new production K19 engines. Oversized liners have dimensions that are oversized with respect to the standard liner.

For example, 20/20 OS means the upper press fit outside diameter (1) is 0.020 inch larger than the standard liner and the lower press fit outside diameter is 0.020 in larger than the standard liner.

The specification for the liner flange thickness on all K19 liners, standard and oversized is the same, 13.360 to 13.386 mm [0.526 to 0.527 in].



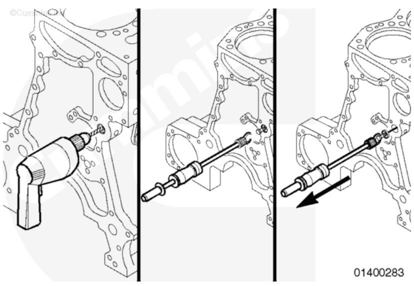
The K38 and K50 liner is referred to as the liner used to in new production K38 and K50 engines. The K38 and K50 production liner upper press fit outside diameter is normally 2.16 mm [0.085 in] larger and the lower press fit outside diameter is normally 1.65 mm [0.065 in] larger than the respective outside diameter on the standard K19 liner.

The K38 and K50 liner can be described as 85/65 oversized with respect to the K19. The use of K38 and K50 oversized upper press fit liners to repair K19 counterbores is **not** recommended.

The use of standard K38 and K50 liners in a properly machined K19 block is authorized.

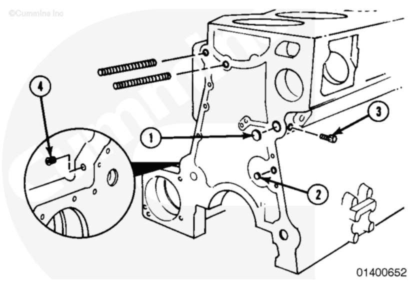
Preparatory Steps

Use a drill, a sheet metal screw, and the listed parts from the light duty puller kit, Part Number 3375784:



- Slide hammer
- Hook.

Remove the listed plugs:

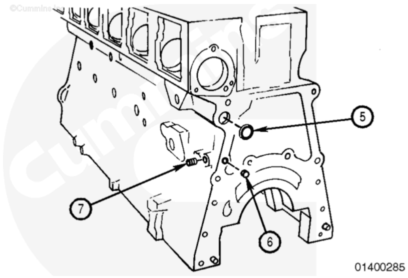


- (1) Main oil passage cup plug
- (2) Piston cooling oil passage cup plug
- (3) Main oil passage pipe plug
- (4) Water pump idler oil passage plug.

Note : Engines equipped with a rear gear drive must not have a plug in the main oil passage.

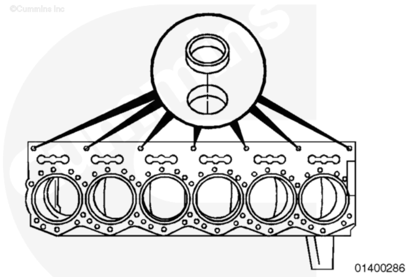
Remove the listed plugs:

- (5) Main oil passage cup plug
- (6) Piston cooling rifle cup plug
- (7) Piston cooling oil passage pipe plug.



Remove the seven cup plugs from the camshaft oil passages.

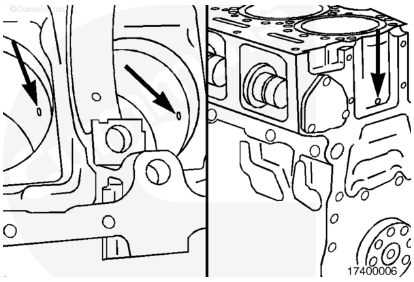
These passages extend through the camshaft bores to the main bearing bores. The passage for the number six main rifle contains a steel tube that has an interference fit with the drilled passage.



To allow for the proper coolant flow and air removal of the cylinder head and engine block on a horizontally mounted K19 engine, a coolant drilling runs the entire length of the cylinder block and passes through the center of each cylinder liner cavity.

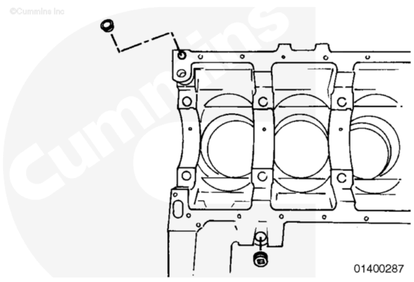
The drilling is plugged by a ¼ pipe plug.

It is **not** necessary to remove the cup plug at the end of the block unless it is leaking.



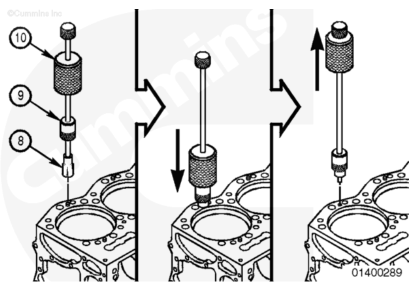
Remove the cup plug from the hydraulic pump idler oil passage.

Remove the pipe plug from the lubricating oil pump oil passage.

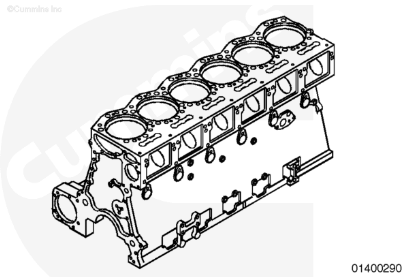


Use dowel pin extractor, Part Number 3163720, or equivalent to perform the listed steps to remove the 12 cylinder head groove pins.

1. Place the split collet (8) over the groove pin.
2. Slide the extractor collar (9) over the split collet.
3. Use the slide hammer (10) to push the extractor collar over the split collet tightly.
4. Use the slide hammer to remove the groove pin.



Clean and Inspect for Reuse



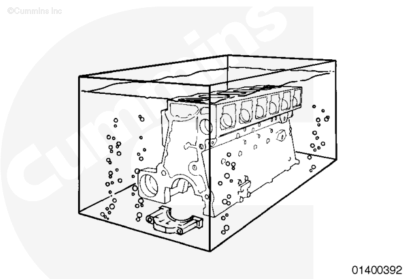
⚠ CAUTION ⚠

To reduce the possibility of leaks and engine damage, do not damage the machine gasket surfaces or the camshaft bushings.

Use a scraper, a wire brush, or hand pad, Part Number 3823258, to clean all the heavy dirt deposits off the cylinder block and clean:

- All gasket surfaces
- All mounting surfaces
- Cylinder liner counterbore ledge and press fit areas
- Cylinder liner packing ring bore
- Top of block
- Main bearing saddle and caps
- All cup plug bores.

Clean all the oil passages with a long handle bottle brush.



⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

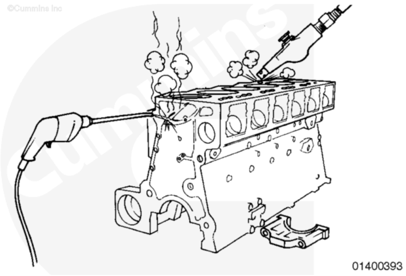
⚠ CAUTION ⚠

A cleaning solution must be used that will not harm camshaft bearings.

Cummins Inc. does **not** recommend any specific cleaning solution. Use the General Cleaning Instructions procedure. Refer to Procedure 204-008 (/qs3/pubsys2/xml/en/procedures/99/99-204-008.html).

Follow the instructions of the manufacturer of the cleaning tank and the manufacturer of the cleaning solution.

Remove the block from the engine stand and place it into a tank of cleaning solution.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

To reduce the possibility of damage to the cylinder block, make sure all the water is removed from the capscrew holes and the oil passages.

Remove the block from the cleaning tank.

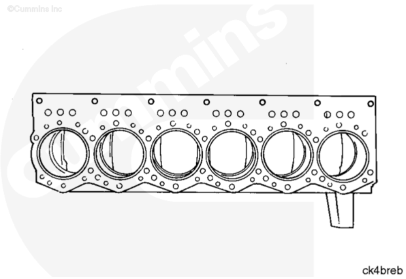
Steam clean the block.

Make sure all the oil passages are clean.

Dry the block with compressed air.

If the cylinder block is **not** going to be used immediately, apply a coating of preservative oil to prevent rust.

Cover the block to prevent dirt from sticking to the oil.



Check the top surface of the block for wear. If fretting damage is present in an area where a head gasket seal ring or a grommet makes contact, the surface **must** be repaired.

Fretting damage in any other area is acceptable if it does **not** change the protrusion measurement of the counterbore or liner.

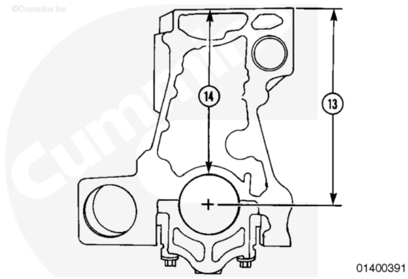
A newly machined surface **must** be flat within 0.05 mm [0.002 in] under a cylinder head. Waves on the surface are acceptable as long as they are **not** more than 0.018 mm [0.0007 in] high, and the high and low points of the waves are **not** closer than 25 mm [1.0 in].

A newly machined surface **must** meet the specifications for block height. The top of the surface of the cylinder liner counterbore **must** be machined.

The parting line of the main bearing cap is **not** at the same height as the centerline of the main bearing bore.

Measure the cylinder block height.

If the checking ring or the centering ring is **not** available, the height of the block can be measured from the top of the bearing saddle (14).



Cylinder Block Height			
	mm		in
(13) Center Line Main Bearing Bore	481.94	MIN	18.974
	482.78	MAX	19.007
(14) Top of Main Bearing Saddle	407.70	MIN	16.051
	408.53	MAX	16.084

If the height of the block from the crankshaft centerline is less than 482.42 mm [18.993 in], a 0.51 mm [0.020 in] oversize head gasket **must** be used.

If the height of the block from the main bearing bore is less than 408.20 mm [16.071 in], a 0.51 mm [0.020 in] oversize head gasket **must** be used.

The height of the block **must not** vary more than 0.05 mm [0.002 in] from end-to-end of the block. If the block height is **not** within specifications, the top surface of the block **must** be machined or the block replaced.

If the top surface of the block is machined, ledge depth of the cylinder liner counterbore **must** be machined.

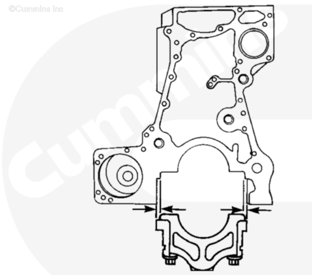
Check the main bearing caps that are loose.

The main bearing cap **must** be repalced if:

- The clearance causes the main bore alignment **not** to be within specifications.
- The clearance between the block and the cap is more than 0.18 mm [0.007 in] on either side of the cap when the cap is installed and tightened to specifications.
- There is fretting or heat damage to the cap.

On new or reconditioned blocks, the main bearing cap is 0.00 to 0.254 mm [0.00 to 0.010 in] larger than the block. Force **must** be used to install the caps.

Service caps do **not** have the bore machined to a final specification. If the cap is machined, use the correct parts of the main bearing boring tool, Part Number ST-1177.



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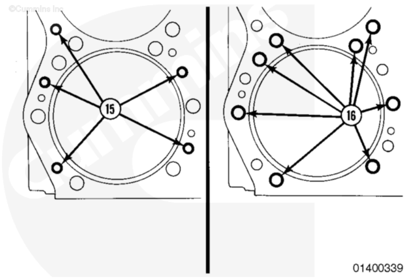
Check the water holes (15). If erosion or pitting is more than 0.08 mm [0.003 in] deep or extends more than 2.4 mm [0.095 in] from the edge of the water hole, the water hole **must** be repaired.

Use the coolant passage repair kit, Part Number 3824047, or the water hole surface repair kit, Part Number 3824066, to repair the water hole.

Check the threads of the capscrew holes (16) for damage. Use a threaded insert if a damaged hole **must** be repaired.



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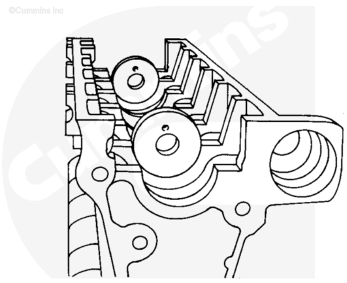
Main Bearing Bore

Place two centering rings, Part Number ST-1177-54, in the number two and number six main bearing locations.



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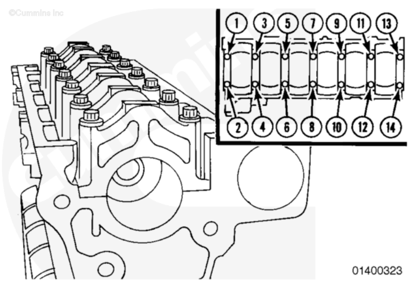
The main bearing caps are numbered one through seven. Install each cap in the correct location. The slot in the cap for the bearing tang **must** be on the same side as the slot in the block.

Position the main bearing caps in the cylinder block.

Install the capscrews, making sure they are positioned correctly.

Use a mallet to drive the caps down until they touch the block.



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If any of the caps do **not** require force during installation, mark the cap to check the side clearance.

Tighten the capscrews for the main bearing caps.

Torque Value:

1. 270 n•m [200 ft-lb]
2. 610 n•m [450 ft-lb]

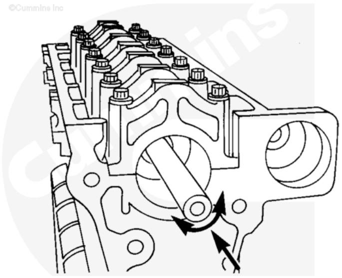
Lubricate the inside diameter of the two centering rings.

Install the alignment/boring bar, Part Number ST-1177-16, in the two centering rings.

The bar **must** turn easily.

If the bar does **not** turn easily, make sure the main bearing caps are installed correctly.

If the main bearing caps appear to be installed correctly, move the centering ring to another bearing location.



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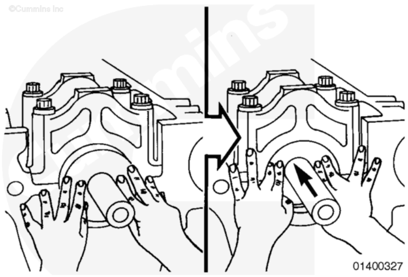
Install the checking ring into the main bearing bore by hand.

If the ring will **not** slide through the main bearing bore, check the bore for burrs.

Remove any burrs in the bore.

If the checking ring still will **not** slide through the bore, the bore is undersized and **must** be repaired. Refer to the Alternate Repair Manual, Bulletin 3379035.



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Use a 0.076 mm [0.003 in] feeler gauge that is **not** more than 13 mm [0.5 in] wide.

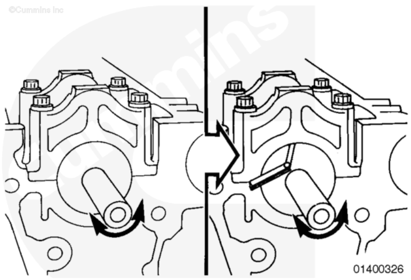
Center the checking ring in the bore.

Attempt to insert the feeler gauge between the checking ring and the bore.

Rotate the gauge in the bore at both sides of the checking ring.

The bore alignment of the main bearing is acceptable if:

- The gauge does **not** enter at any point.
- The gauge will enter but will **not** slide through or around the bore, and the alignment bar will rotate with the gauge inserted.



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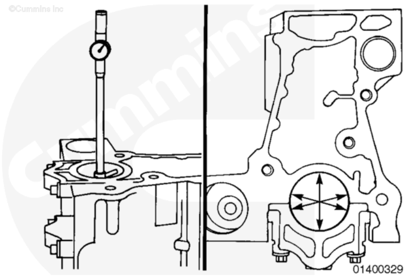
The bore alignment of the main bearing is **not** acceptable if:

- The gauge enters and slides around the bore. This indicates the bore is oversize and **must** be repaired.
- The gauge will enter on one side **only**, but can slide around the bore. This indicates the bore is tapered and **must** be repaired.

If the tools to check the main bearing bore alignment are **not** available, use a dial bore indicator.



Measure the inside diameter in the three positions illustrated in the graphic. The inside diameter **must** be completely round within 0.013 mm [0.0005 in].

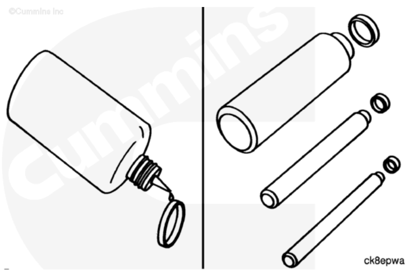
Note : Support the rear portion of the block on a flat surface to obtain the most accurate measurement of the inside diameter.

Main Bearing Bore I.D. (Capscrews Torqued to Specification)		
mm		in
148.44	MIN	5.8444
148.50	MAX	5.8465

Assemble

⚠ CAUTION ⚠

Do not install any cup plugs or pipe plugs in the block until the inspect and any necessary repair procedures are completed. This will prevent dirt from being trapped in any oil passages.



Apply a thin coat of pipe Loctite sealant, Part Number 3375068, or equivalent, to the pipe or cup plugs.

Install the pipe plugs.

Use cup plug driver, Part Number 3164085 or 3376795, with all cup plug drivers to install the cup plugs.

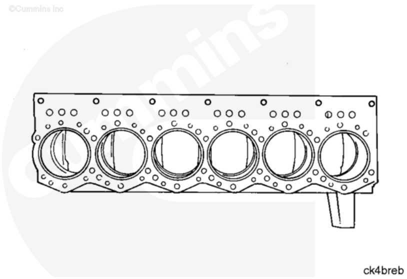
Cup Plug Driver Part Number	Cup Plug Size
3822372	0.375 in

Cup Plug Driver Part Number	Cup Plug Size
3376793	0.500 in
3376813	0.875 in
3376812	1.125 in
3376814	1.625 in

Crack Detection Test

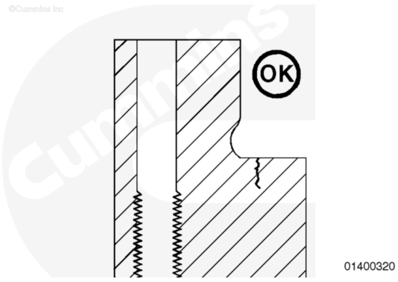
⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



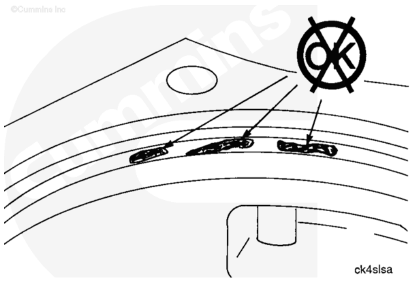
Clean the counterbore area with safety solvent.

Metallurgical analysis of cross sections of counterbores having circumferential cracks has revealed the cracks are surface-initiated on the top of the counterbore ledge, and normally do **not** propagate vertically through the counterbore ledge into the coolant passage around the cylinder liner.



Pitting on the liner seat is **not** acceptable. The illustration in the graphic is an example of pitting in a damaged area.

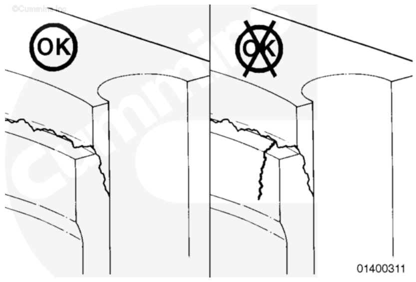
This block requires machining before it can be used. Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/18/18-001-028-tr.html) for the machine depth for the seal ring.



Check the counterbore ledge for cracks with crack detection kit, Part Number 3375432, or equivalent.

Circumferential cracks of the counterbore ledge are acceptable if the cracks do **not** extend to or over the edge of the ledge as shown in the illustration.

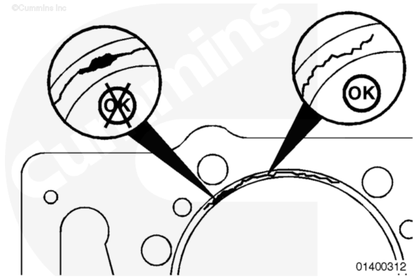
Circumferential cracks in the radius are acceptable if they do **not** extend more than 90 degrees around the radius.



It is **not** necessary to machine the block in an effort to remove acceptable cracks.

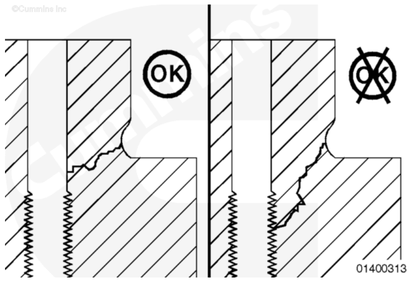
If cracks that are **not** acceptable are found during the initial inspection, the counterbore **must** be machined.

If a crack that is **not** acceptable remains after the machining repair procedure is completed, the block is **not** acceptable for use.



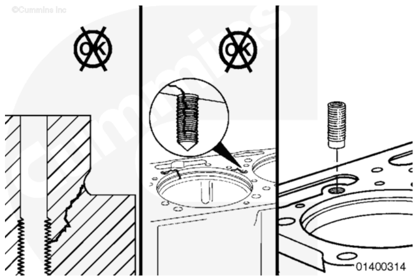
Check the capscrew holes for cracks.

Cracks that extend from the counter bore wall to the capscrew hole are acceptable for use **only** if they do **not** extend into the threaded portion of the hole.



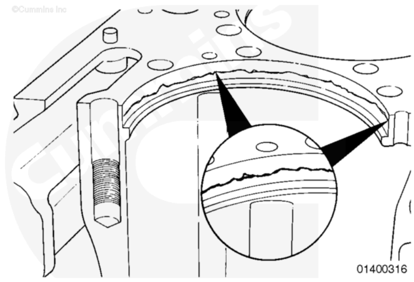
Cracks that extend into the threaded portion of the capscrew hole require repair with a blind-end thread insert.

Use the thread salvage kit, Part Number 3164021 or 3376208, to repair the insert.



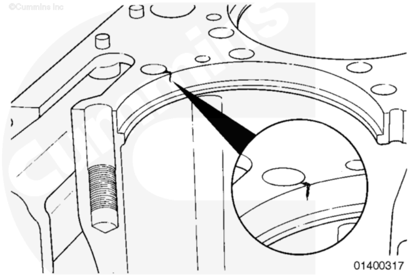
Check for cracks running horizontally around the vertical wall of the counterbore.

All coolant passages that are closest to the bore **must** be repaired with coolant passage threaded inserts.

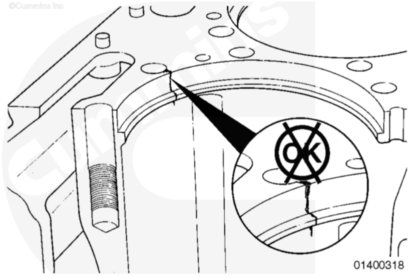


Check for cracks that run vertically to a coolant passage or a capscrew hole.

Those passages **must** be repaired with coolant passage threaded inserts.

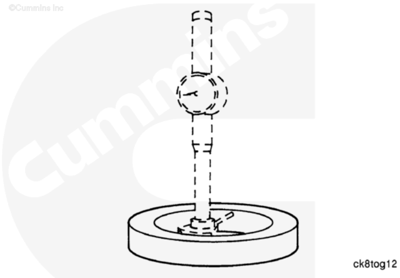


Cylinder block with vertical cracks that extend from a coolant passage down over the counterbore ledge can **not** be repaired.



Measure

Two dial bore gauge setting rings have been released to support salvage of the lower press fit bore in a K19 cylinder block.



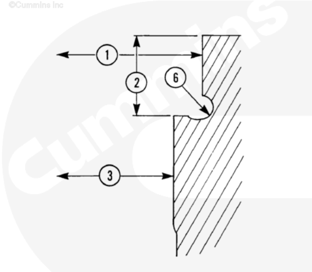
Both rings have steps:

- Standard/0.020 - The inside diameter of the bores in the gauge are:
- Small inside diameter (Standard K19) - 180.11 mm [7.091 in]
- Large inside diameter (0.020 oversized) - 180.62 mm [7.111 in]
- 0.065/0.85 - The inside diameter of the bores in the gauge are:
- Small inside diameter (0.065 oversized) - 181.76 mm [7.156 in]
- Large inside diameter (0.085 oversized) - 182.27 mm [7.176 in]

The 0.065 oversize ring for the K19 is the same as the new production lower press fit ring on the K38 and K50 engines.

Both of the setting rings are included in counterbore salvage kit, Part Number 3824119.

The following lower press fit bore specifications apply to the thick flange, thin flange factory modified, ex-thin, and thin flange type blocks.



ck4brgb

(3) New Block Lower Press Fit Specifications		
mm		in
180.09	MIN	7.090
180.14	MAX	7.092

New Block Maximum Out of Round		
mm		in
0.05	MAX	0.002

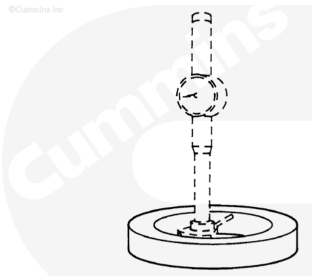
In Service Lower Press Fit Specifications		
mm		in
180.09	MIN	7.090
180.16	MAX	7.093

In Service Maximum Out of Round		
mm		in
0.08	MAX	0.003

There **must** be a minimum of 0.025 mm [0.001 in] press fit all around between the lower press fit outside diameter of the liner and the lower press fit inside diameter of the block. Often, the new standard K19 liners can be sorted by measuring to find a liner that will provide adequate press fit without machining the lower press fit inside diameter of the block.

If the lower press fit inside diameter is **not** within specifications, it **must** be machined to accept either an oversized lower press fit K19 liner or K38 or K50 liner.

Use a dial bore gauge, Part Number 3376619 or 3375072, or equivalent. Other measuring devices such as inside diameter micrometers or calipers are **not** as accurate as a dial bore gauge and can cause unnecessary machining of cylinder blocks.

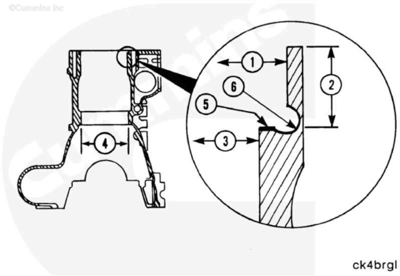


ck8tog12

Ring gauge, Part Number 3376831, has an inside diameter of 188.214 mm [7.410 in]. Use this gauge when setting up the dial bore gauge to measure thick flange, factory modified thin flange, and ex-thin flange blocks that have **not** been machined for oversized upper press fit K19 liners.

Ring gauge, Part Number 3376832, has an inside diameter of 190.335 mm [7.4935 in]. Use this gauge when setting up the dial bore gauge to measure blocks that have been machined for 83/Standard oversized upper press fit liners or K38 and K50 standard liners.

The specification for the upper press fit inside diameter (1) is dependent on whether or **not** the block has been machined for oversized upper press fit liners.



For blocks in service, the upper press fit inside diameter **must** be **not** more than the 0.076 mm [0.003 in]. Smaller than 0.076 mm [0.003 in], larger than the liner flange outside diameter on the liner that is to be installed in that bore. This bore can be as much as 0.15 mm [0.006 in] out of round.

When machining the block to accept the oversized lower press fit or K38 and K50 liners, the inside diameter (1) needs to be the same size as the 0.076 mm [0.003 in], larger than the liner to be installed in that bore.

Use a depth micrometer. When measuring a ex-thin flange block, make sure the end of the gauge tip is **not** contacting the ledge radius.

Measure the depth at four points.

The four measures **must not** vary more than 0.25 mm [0.001 in].

The maximum depth for any K19 block, except the counterbore ring type, is 158.8 mm [0.625 in]. This maximum depth also applies if the block deck height is at the minimum specifications.

Counterbore Repair Matrix - Thick Flange and Factory Modified Type Blocks			
Liner Removed	**Lower Press Fit in Specification**	**Upper Press Fit in Specification**	**Recommended Counterbore Repair**
Standard	Yes	Yes	Machine depth for seal ring, use standard liner
Standard	Yes	No	Machine for 10/10 liner and seal ring

Counterbore Repair Matrix - Thick Flange and Factory Modified Type Blocks			
Liner Removed	**Lower Press Fit in Specification**	**Upper Press Fit in Specification**	**Recommended Counterbore Repair**
Standard	No	Yes	Machine for 10/10 liner and seal ring
Standard	No	No	Machine for 10/10 liner and seal ring
20/Standard	Yes	Yes	Machine depth for seal ring, Use 20/standard oversize upper press fit liner
20/Standard	Yes	No	Machine for 40/standard liner and seal ring
20/Standard	No	Yes	Machine for 20/20 liner and seal ring
20/Standard	No	No	Machine for 60/20 liner and seal ring
40/Standard	Yes	Yes	Machine depth for seal ring, use 40/standard oversize upper press fit liner
40/Standard	Yes	No	Machine for 40/standard liner and seal ring
40/Standard	No	Yes	Machine for 60/20 liner and seal ring
40/Standard	No	No	Machine for 60/20 liner and seal ring
60/Standard	Yes	Yes	Machine depth for seal ring, use 60 standard oversize upper press fit liner
60/Standard	Yes	No	Machine for K38 and K50 liner and seal ring
60/Standard	No	Yes	Machine for 60/20 liner and seal ring
60/Standard	No	No	Machine for K38 and K50 liner and seal ring

Counterbore Repair Matrix - Thick Flange and Factory Modified Type Blocks			
Liner Removed	**Lower Press Fit in Specification**	**Upper Press Fit in Specification**	**Recommended Counterbore Repair**
83/Standard	Yes	Yes	Machine depth for seal ring, use 83/standard oversize upper press fit liner
83/Standard	Yes	No	None, upper press fit already over K38 and K50 standard
83/Standard	No	Yes	Machine for K38 and K50 liner and seal ring
83/Standard	No	No	None, bores already over K38 and K50 standard
95/Standard	Yes	Yes	Machine depth for seal ring, Use 95/standard oversize upper press fit liner
95/Standard	Yes	No	None, upper press fit already over K38 and K50 standard
95/Standard	No	Yes	None, upper press fit already over K38 and K50 standard
95/Standard	No	No	None, bores already over K38 and K50 standard
10/10	Yes	Yes	Machine depth for thicker seal ring, use 10/10 liner
10/10	Yes	No	Machine for 20/20 liner and seal ring
10/10	No	Yes	Machine for 20/20 liner and seal ring
10/10	No	No	Machine for 20/20 liner and seal ring
20/20	Yes	Yes	Machine depth for thicker seal ring, use 20/20 liner

Counterbore Repair Matrix - Thick Flange and Factory Modified Type Blocks			
Liner Removed	**Lower Press Fit in Specification**	**Upper Press Fit in Specification**	**Recommended Counterbore Repair**
20/20	Yes	No	Machine for 60/20 and thicker seal ring
20/20	No	Yes	Machine for K38 and K50 standard and thicker seal ring
20/20	No	No	Machine for K38 and K50 standard and thicker seal ring
60/20	Yes	Yes	Machine for thicker seal ring, use 60/20 liner
60/20	Yes	No	Machine for K38 and K50 standard and thicker seal ring
60/20	No	Yes	Machine for K38 and K50 standard and thicker seal ring
60/20	No	No	Machine for K38 and K50 standard and thicker seal ring
K38 and K50	Yes	Yes	Machine for thicker seal ring, use K38 and K50 standard liner
K38 and K50	Yes	No	None, upper press fit already over K38 and K50 standard
K38 and K50	No	Yes	None, upper press fit already over K38 and K50 standard
K38 and K50	No	No	None, bores already over K38 and K50 standard

Counterbore Repair Matrix - Ex-Thin Type Blocks			
Liner Removed	**Lower Press Fit in Specification**	**Upper Press Fit in Specification**	**Recommended Counterbore Repair**
60/Standard	Yes	Yes	Machine depth for seal ring, use 60/standard oversize upper press fit liner
60/Standard	Yes	No	Machine for K38 and K50 liner and seal ring
60/Standard	No	Yes	Machine for 60/20 oversize and seal ring
60/Standard	No	No	Machine for K38 and K50 liner and seal ring
83/Standard	Yes	Yes	Machine depth for seal ring, use 83/standard of upper press fit liner
83/Standard	Yes	No	None, upper press fit already over K38 and K50 liner
83/Standard	No	Yes	Machine for K38 and K50 liner and seal ring
83/Standard	No	No	None, bores already over K38 and K50 standard
95/Standard	Yes	Yes	Machine depth for seal ring, use 95/standard oversize upper press fit liner
95/Standard	Yes	No	None, upper press fit already over K38 and K50 standard
95/Standard	No	Yes	None, upper press fit already over K38 and K50 standard
95/Standard	No	No	None, bores already over K38 and K50 standard

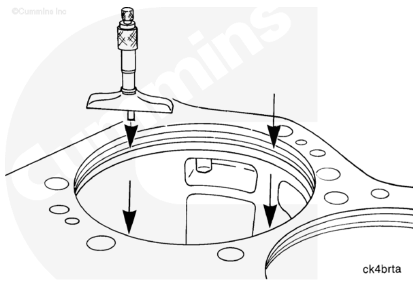
Counterbore Repair Matrix - Ex-Thin Type Blocks			
Liner Removed	**Lower Press Fit in Specification**	**Upper Press Fit in Specification**	**Recommended Counterbore Repair**
60/20	Yes	Yes	Machine for thicker seal ring, use 60/20 liner
60/20	Yes	No	Machine for K38 and K50 standard and thicker seal ring
60/20	No	Yes	Machine for K38 and K50 standard and thicker seal ring
60/20	No	No	Machine for K38 and K50 standard and thicker seal ring
K38 and K50	Yes	Yes	Machine for thicker seal ring, use K38 and K50 standard liner
K38 and K50	Yes	No	None, upper press fit already over K38 and K50 standard
K38 and K50	No	Yes	None, lower press fit over K38 and K50 standard
K38 and K50	No	No	None, bores already over K38 and K50 standard

Counterbore Repair Matrix - Thin Flange Type Blocks			
Liner Removed	**Lower Press Fit in Specification**	**Upper Press Fit in Specification**	**Recommended Counterbore Repair**
Thin flange	Yes	Yes	Machine depth for seal ring, use thin flange liner
Thin flange	Yes	No	Machine for 60/standard liner and seal ring
Thin flange	No	Yes	Machine for 60/20 liner and seal ring
Thin flange	No	No	Machine for 60/20 liner and seal ring

Counterbore Machining Specifications				
Liner Upper Press Fit/Lower Press Fit	Upper Press Fit Bore Inside Diameter		Lower Press Fit Bore Inside Diameter	
Oversize/Oversize	**Minimum mm [in]**	**Maximum mm [in]**	**Minimum mm [in]**	**Maximum mm [in]**
Standard/Standard	188.16 [7.408]	188.32 [7.414] [1]	180.09 [7.090]	180.16 [7.093] [2]
10/10	188.44 [7.419]	188.57 [7.424]	180.34 [7.100]	180.42 [7.103]
20/20	188.75 [7.431]	188.82 [7.434]	180.59 [7.110]	180.64 [7.112]
60/20	189.76 [7.471]	189.84 [7.474]	180.59 [7.110]	180.64 [7.112]
K38/K50	190.37 [7.495]	190.45 [7.498]	181.74 [7.155]	181.79 [7.157]
20/Standard	188.67 [7.428]	188.72 [7.430]	180.09 [7.090]	180.16 [7.093] [2]
40/Standard	189.18 [7.448]	189.23 [7.450]	180.09 [7.090]	180.16 [7.093] [2]
60/Standard	189.69 [7.468]	189.74 [7.470]	180.09 [7.090]	180.16 [7.093] [2]
83/Standard	190.27 [7.491]	190.32 [7.493]	180.09 [7.090]	180.16 [7.093] [2]
95/Standard	190.58 [7.503]	190.63 [7.505]	180.09 [7.090]	180.16 [7.093] [2]
Thin flange/Standard	188.01 [7.402]	188.09 [7.405] [1]	180.09 [7.090]	180.16 [7.093] [2]
[1] Maximum inside diameter blocks in service.				
[2] Maximum inside diameter blocks in service. **Must** be a minimum of 0.03 mm [0.001 in] press fit between liner and block.				

Measure the counterbore depth at the four locations illustrated in the graphic with a depth micrometer.

Make sure the micrometer contacts the flat surface of the ledge. It **must not** touch the radius.



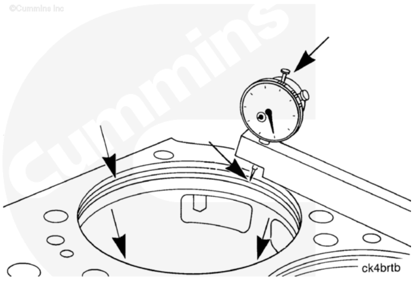
Counterbore Depth		
mm		in
13.754	MIN	0.5415
13.805	MAX	0.5435

The four measurements **must not** vary more than 0.25 mm [0.001 in]. If the measurements exceed the specifications, the counterbore ledge **must** be machined.

Make sure the indicator does **not** contact the counterbore radius on a block that does **not** have a double undercut.

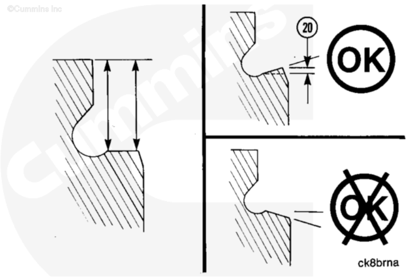
Use depth gauge assembly, Part Number 3164438 or 3823495, or equivalent, to measure the angle of the counterbore ledge at four places on the counterbore circumference.

The measurement of the ledge depth **must** be as near to the counterbore radius as possible, and as near to the counterbore edge as possible.



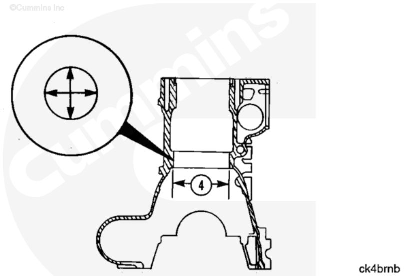
The angle (12) of the counterbore ledge is acceptable if the measurement near the counterbore is the same or no more than 0.36 mm [0.0014 in] shorter than the measurement near the counterbore radius.

If the measurement near the counterbore ledge is greater than the measurement near the counterbore radius, the ledge **must** be machined.



Measure the chamfer at the top of the packing ring bore. Excessive pitting **must** be repaired.

(4) Packing Ring Bore		
mm		in
177.34	MIN	6.982
177.39	MAX	6.984



If the packing ring bore is **not** within specifications, it **must** be repaired. Refer to the Alternate Repair Manual, Bulletin 3379035.

Last Modified: 02-Jan-2019